dashboard, you see an increased error rate in the app, and to figure out which services contribute to the error, you use metrics–traces correlation to find traces of services causing errors.

- *Root cause analysis*—Suppose there was a recent incident, and you want to figure out what caused it. Metrics telemetry provides you with information about the compute environment (e.g., region), and you can gather version information (e.g., about the libraries used, the service itself, etc.) from logs. To figure out what the causes of an outage were, you can use metrics–logs correlation to determine whether the problem is relevant for certain machine types, service versions, regions.
- *Service performance improvements*—Let's assume you're a developer working on a ticket that calls out a certain number of customers reporting slower execution for an endpoint. The service graph provided by distributed tracing allows you to pinpoint services that, together, handle the request path for said endpoint. To determine which services to improve, you use traces–profiles correlation and compare profiles of service versions you can optimize.

While it's possible to use multiple signal types together based on manual correlation, it's not an effective use of your time. By *manual signal correlation*, I mean starting with a signal type, such as traces, and, based on the time range covered, trying to find, for example, log lines of the services that contribute to the trace and hoping they are somewhat relevant. Hope is not a strategy. We will first discuss signal correlation on a conceptual level in this final chapter. Then, we will see correlation in action, using tooling you are already familiar with. Finally, we will wrap up with some considerations for implementing signal correlation.

11.1 Correlation fundamentals

Let's first define what we mean by *signal correlation* and then look at how it works conceptually. I've based the signal correlation definition on Bartek Płotka's thoughts in "Correlating Signals Efficiently in Modern Observability" (https://www.bwplotka.dev/2021/correlations-exemplars/). I've not only referenced Płotka's work because I know and trust him but also because he served as a tech lead in the CNCF TAG Observability.

> **DEFINITION** *Signal correlation* is the metadata-driven process of connecting different signal types, with the goal of gaining actionable insights faster and/or more accurately. The metadata enables automation of multisignal observability. A correlation path is the transitioning from one signal type to a correlated one.

The correlation stack, shown in figure 11.1, shows the conceptual layers of how signal correlation can be achieved. Remember that we want to automate as much as possible, so the better the layers are integrated, the closer we may come to full automation.